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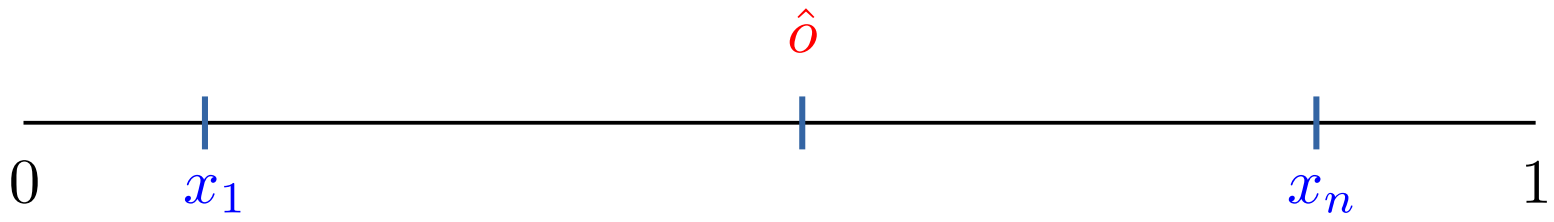
Assumption: $x_1 \leq x_2 \leq \ldots \leq x_n$, and $\hat{o}$ is the prediction

Mechanism $\mathbf{MinMaxP}(x_1, x_n, \hat{o})$: $y = \text{median}\{x_1, x_n, \hat{o}\}$.

Each agent i reports $p_i \in [0, 1]$

Each agent $i's$ utility : $u_i = -|p_i - y|$



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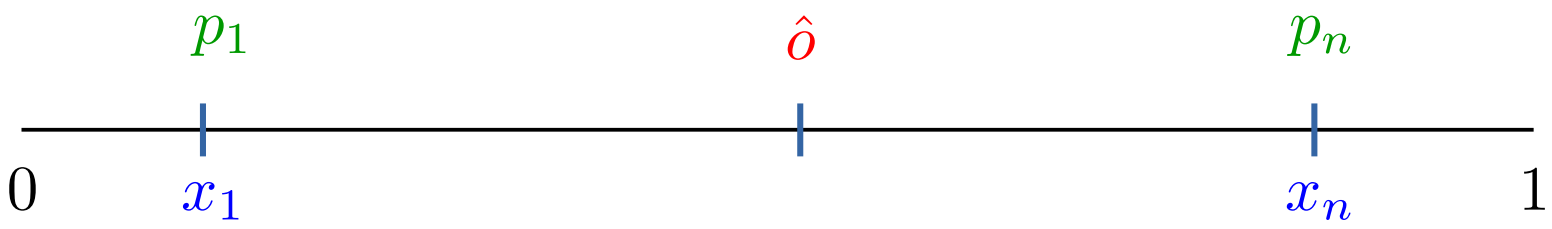
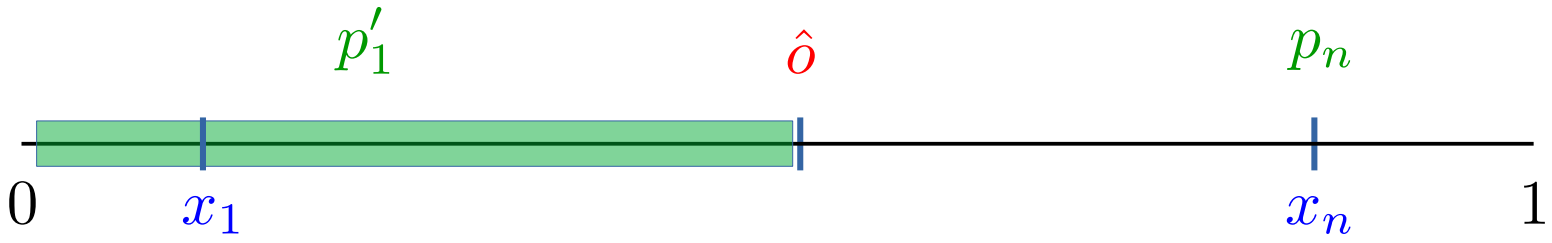
$y = \hat{o}$ if the prediction is accurate and $p_i = x_i \; \forall i$



Suppose agent 1 lies to p_1' and the new outcome is $y = \text{median}(p_1', \hat{o}, p_n)$

Case 1: $p_1' \leq \hat{o}$

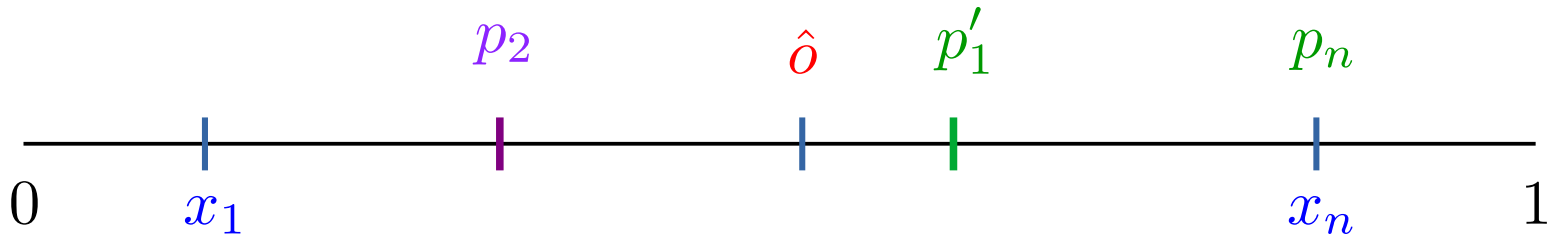
$$y = \text{median}(p_1', \hat{o}, p_n) = \hat{o} \;\Rightarrow\; u_1 \text{ stays the same.}$$



Suppose agent i lies to p_i' and the new outcome is $y = \text{median}(p_1', \hat{o}, p_n)$

Case 2(a): $\hat{o} \leq p_1' \leq p_n$

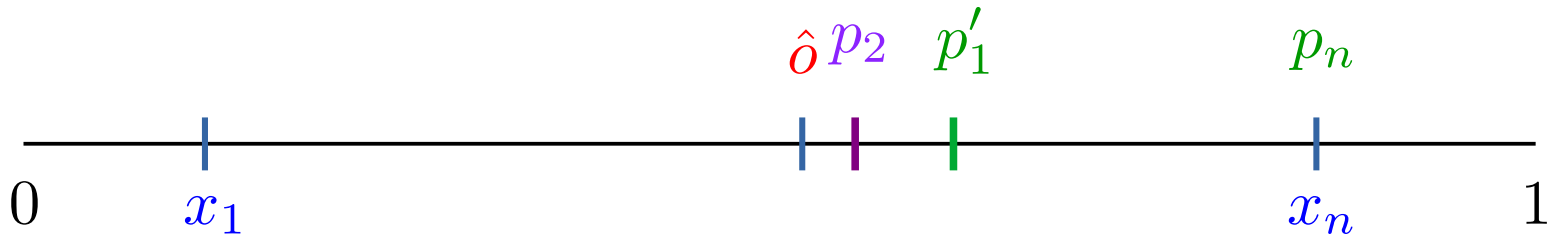
$$y = \text{median}(p_2, \hat{o}, p_n) = \hat{o} \text{ if } p_2 \leq \hat{o} \;\Rightarrow\; u_1 \text{ stays the same.}$$



Suppose agent i lies to p_i' and the new outcome is $y = \text{median}(p_1', \hat{o}, p_n)$

Case 2(b): $\hat{o} \leq p_1' \leq p_n$

$$y = \text{median}(p_2, \hat{o}, p_n) = p_2 \text{ if } p_2 > \hat{o} \;\Rightarrow\; u_1 = -|y - x_1| \downarrow$$



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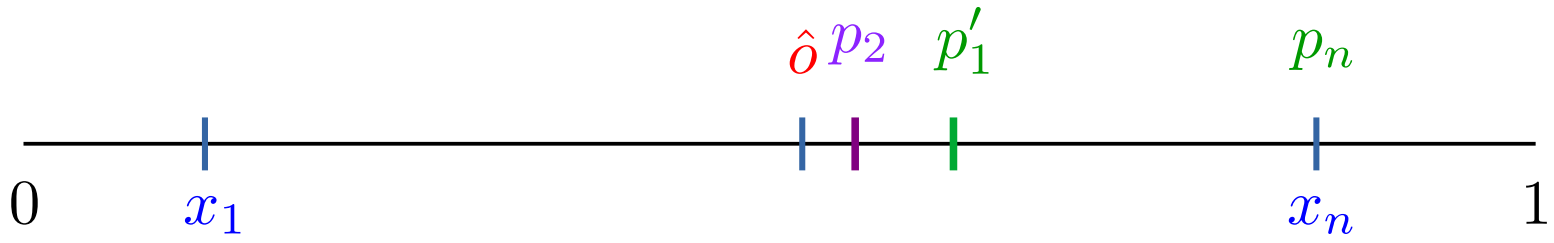
Therefore, agent 1 cannot gain more utility by misreporting.
Similar arguments hold for agent n.



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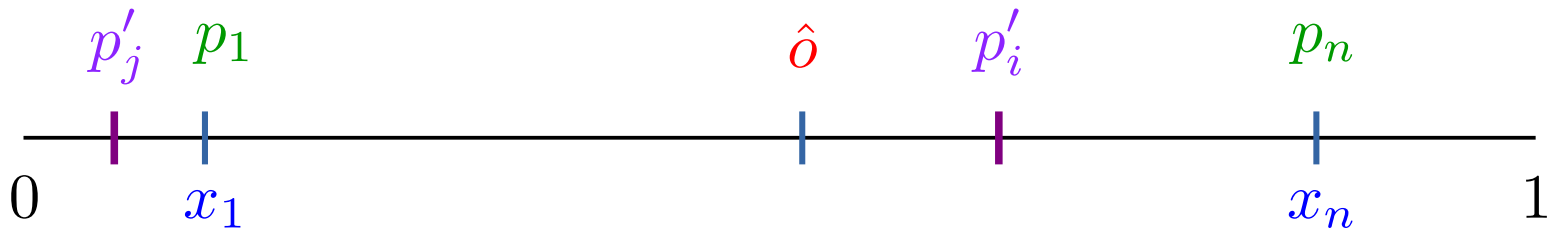
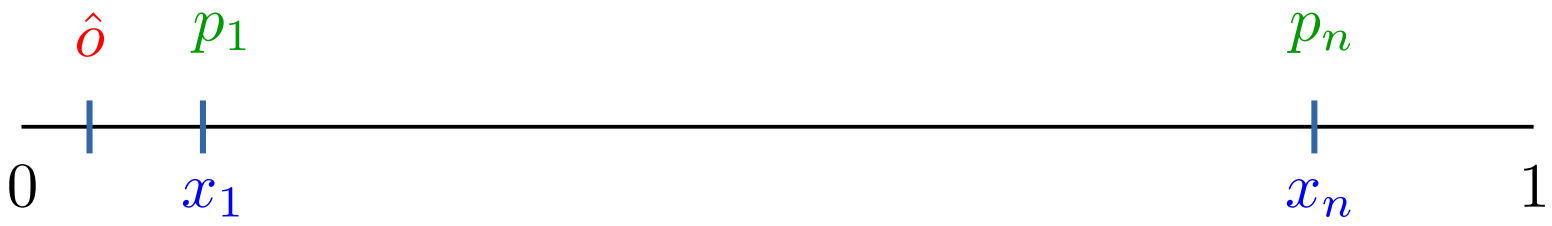
For agent $i, i \in \{2, 3, \ldots, n-1\}$, if $p_1 \leq p_i' \leq p_n$, the outcome y is unchanged. and if $p_i' \leq p_1$ or $p_n < p_i', y$ is still unchanged.



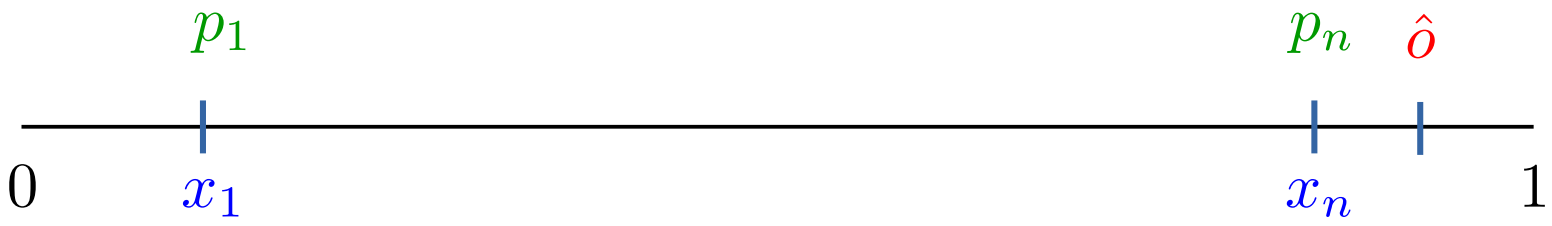
What if the prediction $\hat{o}$ is inaccurate ?

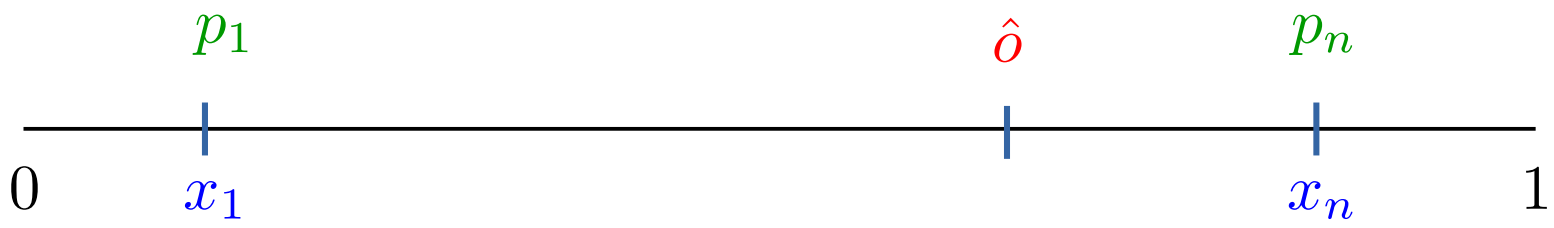
agent 1 is truth-telling:

$$y = \text{median}(p_1, \hat{o}, p_n) = p_1 \;\Rightarrow \text{being truthful is optimal.}$$



Similar case for agent n.



prediction $\hat{o}$ is between p_1 and p_n

$$y = \text{median}(p_1, \hat{o}, p_n) = \hat{o} \text{ which is independent of agents' reports.}$$